

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 08 · LECTURE NOTES

Combinatorial Auctions

Where computation and incentives pull against each other.

MODULE OUTCOMES

- Explain why bundles make the problem hard.
- State the winner determination problem and its complexity.
- Explain the tension between approximation and truthfulness.
- Explain restricted valuation classes and what they buy.
- Explain why practical designs use indirect mechanisms.

1 Complements and substitutes

A BIDDER'S VALUE FOR A BUNDLE NEED NOT BE THE SUM OF ITS PARTS, WHICH IS WHAT MAKES THIS A SEPARATE SUBJECT

Complements: a pair of adjacent spectrum licences, or an outbound and a return flight — worth far more together than separately, so winning one of them alone can be worthless or actively harmful, since you paid for something unusable.

Substitutes: two interchangeable delivery slots, or two equivalent servers — worth much less together than the sum of their individual values, so winning both is money wasted.

Which means running separate auctions exposes bidders to exposure risk: bid aggressively for a complement and you may end up winning only half of what you needed, which makes rational bidders bid timidly and makes the outcome inefficient.

So bundles have to be expressible, and a general valuation over m items is a function on 2^m bundles — which cannot even be written down for any realistic m — and that is §4's elicitation problem arriving before §2's computational one, and is arguably the more fundamental of the two.

2 Winner determination

FINDING THE VALUE-MAXIMISING ALLOCATION IS NP-HARD, AND VCG NEEDS IT SOLVED EXACTLY

Winner determination is essentially a weighted set packing problem — NP-hard, and hard to approximate within any reasonable factor in the general case (CSCE 629 Module 12's reductions, CSCE 637 Module 11's inapproximability results).

And VCG requires the exact optimum, computed $n + 1$ times — once overall and once with each agent removed (Module 06 §3) — **so VCG is simply unavailable at any realistic scale**, which is a computational objection to an economic result and is what algorithmic game theory was founded to notice.

But swapping in an approximation algorithm breaks the truthfulness proof — which used exact optimality in an essential way (the agent's utility equalled the total welfare up to a constant only because the allocation was optimal) — **and the resulting mechanism is manipulable**, sometimes severely.

So the field's central question became: which approximation algorithms can be made truthful, and at what loss in the approximation ratio? — which is a genuinely new kind of algorithmic question, and one that has produced techniques (monotone allocation rules, critical-value payments) with no classical counterpart.

3 What can be done

- 1 RESTRICT THE VALUATIONS
 single-minded bidders (each wants exactly one bundle), gross substitutes, submodular valuations.
 Under these restrictions, truthful approximations exist and are sometimes essentially optimal.
 COST: the restriction may not hold, and checking whether it does may be hard.
- 2 ACCEPT A WEAKER INCENTIVE PROPERTY
 truthful in expectation, or truthful under a Bayesian model, or approximately truthful
 COST: the guarantee now assumes something about beliefs (Module 02 section 2), which is exactly what dominant strategies avoided
- 3 USE AN INDIRECT MECHANISM
 ascending auctions, in which bidders respond to posted prices rather than reporting a whole valuation function
 COST: no clean truthfulness theorem, and strategic behaviour is possible
 GAIN: bidders reveal only what the prices require, which is the real reason they are used (section 4)

AND THE DEPLOYED SPECTRUM AUCTIONS USE ROUTE 3.

The deployed spectrum auctions use indirect ascending mechanisms, not VCG — which is the field's most instructive practical fact, given that those auctions were designed by people who knew the theory extremely well and had every reason to use the optimal mechanism if it had been usable. The reasons are §4's, and they are not computational.

4 Why indirect wins in practice

- Bidders never have to state a full valuation function — which they could not compute and would not want to disclose — they respond only to the prices they actually face, which is a vastly smaller demand.
- Which addresses the preference elicitation problem: determining a bidder's own value for a particular bundle may itself be expensive — requiring engineering studies, market analysis, or negotiation — and prices tell them which bundles are worth the trouble of evaluating.
- And it reveals less — commercially sensitive valuations stay private except insofar as the prices force them out — which bidders care about a great deal, and which a direct revelation mechanism ignores entirely.

- **Plus it is comprehensible — participants understand an ascending price — and a mechanism nobody understands is not truthful in practice**, whatever the theorem says (Module 07 §4's shading, and Module 13 §1).
- **At the cost of having no clean truthfulness theorem, which is a trade the designers made deliberately and defended in writing. Determining your own value for a bundle may itself be expensive — which is the preference elicitation problem, and the direct-revelation framing assumes it away entirely.**

AND THE LESSON GENERALISES PAST AUCTIONS

The theoretically dominant mechanism lost to practical considerations that the model omitted — computability, elicitation cost, privacy, and comprehensibility — all four of which are real and none of which appears in the VCG theorem.

Which is not a failure of the theory: the theory identified the benchmark, and the practical designs are evaluated against it — you cannot say how much a design gives up without knowing what the optimum was.

And it is the shape of this whole semester — the model comes from outside, and the parts the model omits are where the engineering lives (CSCE 640 Module 13 §3's baselines, CSCE 628 Module 13 §4's external model).

So when you design one: state the benchmark, state what you gave up, and state why — which is Module 13 §3's claim form and Project 2's requirement.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Cramton, Shoham & Steinberg — Combinatorial Auctions	The whole module — the standard collection, with chapters on each of §3's routes.
Lehmann, O'Callaghan & Shoham — Truth revelation in approximately efficient combinatorial auctions (free)	§§2 and 3 — the single-minded case, and the monotonicity technique that makes approximation truthful.
Nisan et al., chapters 11 and 12 (free PDF)	§§2 and 3 — computationally efficient mechanisms, with the tension developed carefully.
Milgrom — Putting Auction Theory to Work	§4 — by a designer of the spectrum auctions, on why the practical choices were made. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Construct a complements example** and compute the exposure risk.
- 2 **Construct a substitutes example.**
- 3 **Count the numbers needed** to express a general valuation over 20 items.
- 4 **Reduce set packing to winner determination**, or the other way.
- 5 **Implement VCG** for a small combinatorial instance and time it as m grows.
- 6 **Replace the exact solver with a greedy one** and find a profitable misreport.
- 7 **Implement the single-minded truthful approximation** and verify monotonicity.
- 8 **Run an ascending auction** on the same instance and compare the outcome.
- 9 **Estimate the elicitation cost** for a real bundle valuation you can imagine.
- 10 **Name the four practical considerations** and find a system where each binds.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define complements and substitutes, with an example of each.
- 2 What is exposure risk?
- 3 How large is a general valuation function?
- 4 What is winner determination, and how hard is it?
- 5 Why does approximation break VCG's truthfulness?
- 6 State the field's central question.
- 7 Give the three routes out, with their costs.
- 8 Which route do the spectrum auctions use?
- 9 Give four reasons indirect mechanisms win in practice.
- 10 State the general lesson.